

ER Queue Management System

Doubly Linked List Implementation

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Problem Statement

Emergency rooms need efficient queue management where critical patients get immediate priority while normal patients are treated in arrival order, with support for dynamic insertion and removal.

Proposed Solution

Doubly Linked List with bidirectional pointers enabling: critical patient insertion at front, normal patients at end, flexible positioning, and efficient removal of treated patients.

Operations Visualization

`insertAtEnd(101)`

101

`insertAtEnd(102)`

101

↔

102

`insertAtBeginning(200)`

200

↔

101

↔

102

`insertAtPosition(150, 2)`

200

↔

150

↔

101

↔

102

`deleteFromBeginning()`

150

↔

101

↔

102

`insertAtEnd(300)`

150

↔

101

↔

102

↔

300